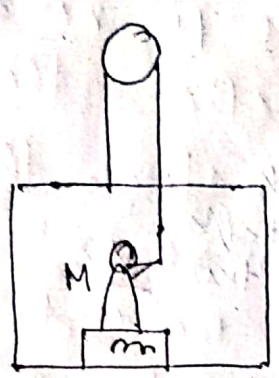
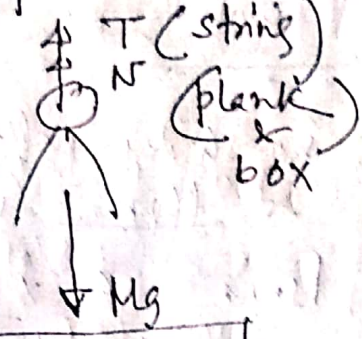
NLM (JEE) Solution

(1)

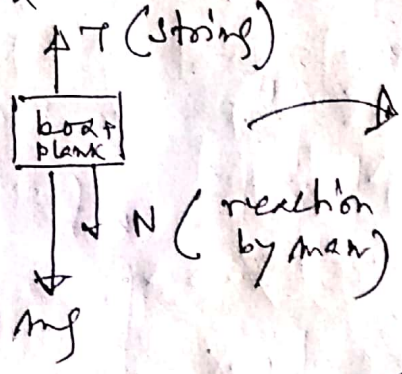


f.b.d of man



$$\boxed{T + N = Mg \quad -(1)}$$

f.b.d of plank & box



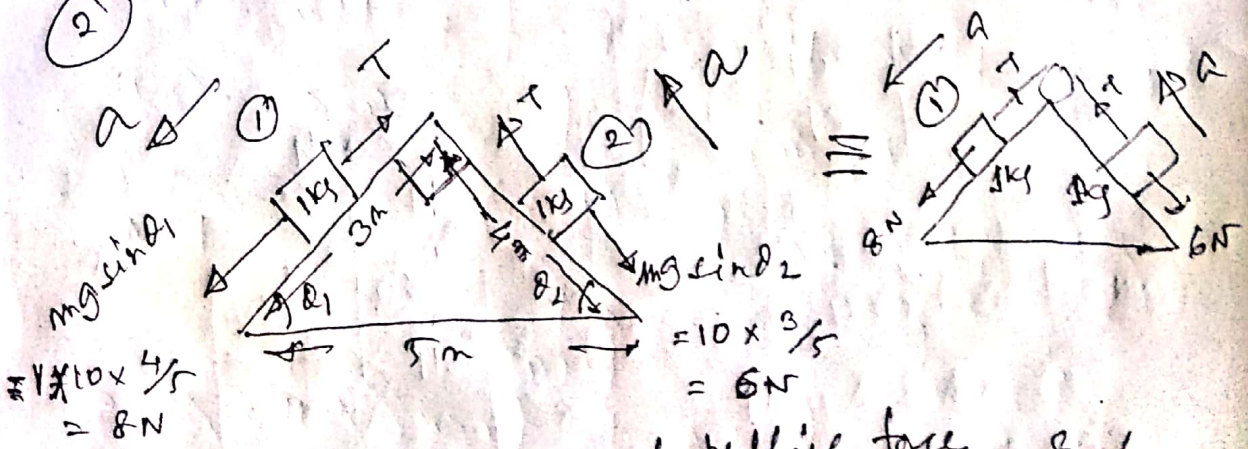
$$T = N + mg$$

$$\boxed{T - N = mg} \quad -(2)$$

Solving (1) & (2) we get

$$\boxed{T = \frac{(M+m)g}{2}}, \ \& \ \boxed{N = \frac{(M-m)g}{2}}$$

(2)



$$a_{system} = \frac{\text{Net pulling force}}{\text{total mass}} = \frac{8-6}{1+1}$$

$$= \frac{2}{2} = 1\,m/s^2 = \frac{g}{10}\,m/s^2.$$

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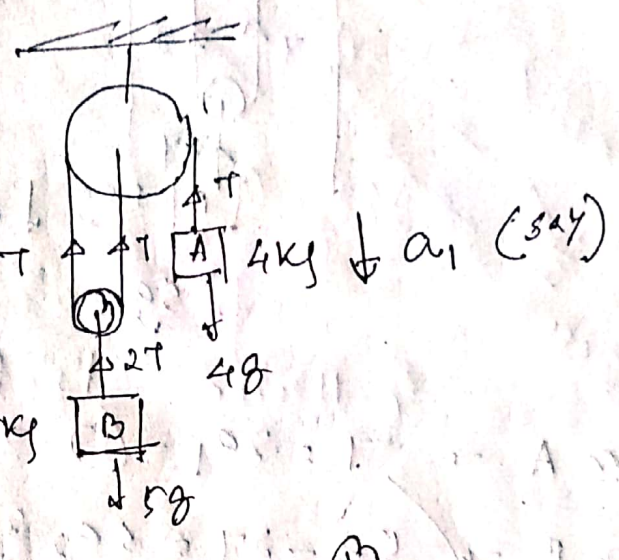
Problem.

(3) (4) 5 (c) (6) are solved in NLM JEE VIDEO CLASS. Problem (11), (13), (18) also solved in this class

5) a)



for (A) $4g - T = 4a_1$ — (1)

for (B) $2T - 5g = 5a_2$ — (2)

[rel. b/w a_1 & a_2 ⇒ $-Ta_1 + 2Ta_2 = 0$

$\sum \vec{T} \cdot \vec{a} = 0$ ⇒ $a_1 = 2a_2$

If $a_2 = a$ then $a_1 = 2a$]

(1) → $4g - T = 4 \times 2a$

$4g - T = 8a$ — (1)

(2) → $2T - 5g = 5a$ — (2)

(1) × 2 + (2)

$8g - \cancel{2T} = 16a$

$\cancel{2T} - 5g = 5a$

$3g = 21a$

$a = \frac{g}{7} = a_2$

& $a_1 = 2a = \frac{2g}{7}$ downwards

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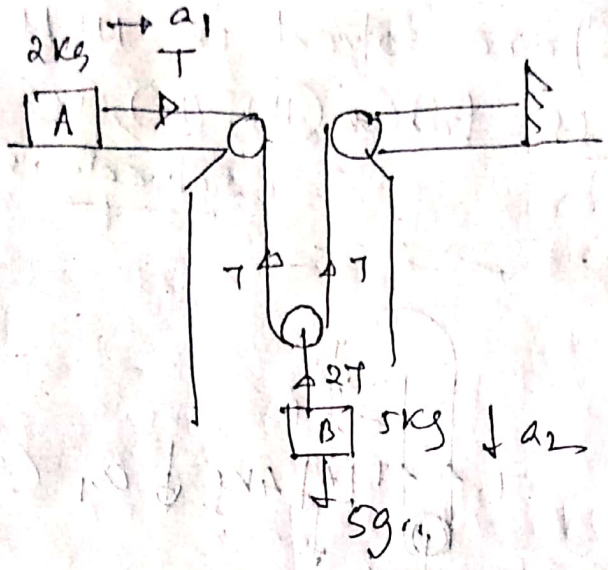
5) (b)



a_1, a_2 rel.

$\Sigma \vec{T} \cdot \vec{a} = 0$

$Ta_1 - 2Ta_2 = 0$

$a_1 = 2a_2$

if $a_2 = a$

then $a_1 = 2a$.

for A → $T = 2a_1$ — (1)

for B → $5g - 2T = 5a_2$ — (2)

$T = 2 \times 2a$

$T = 4a$ — (1)

& $5g - 2T = 5a$ — (2)

(1) × 2 + (2), $\cancel{2T} = 8a$

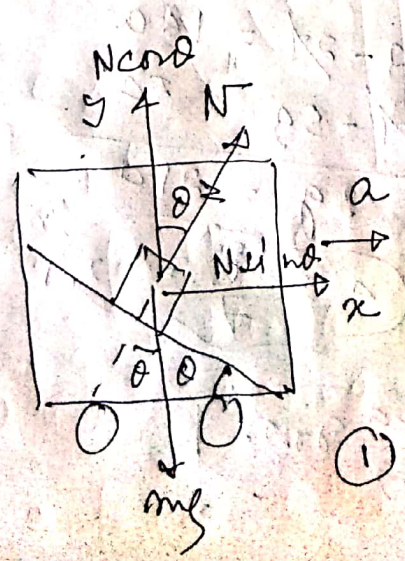
$5g - \cancel{2T} = 5a$

$5g = 13a \Rightarrow \boxed{a \sim a_2 = \frac{5g}{13}}$

$a_1 = 2a$

$\boxed{a_1 = \frac{10g}{13}}$.

7



w.r.t ground frame (inertial frame)

$N \sin\theta = ma$ — (1)

$N \cos\theta = mg$ — (2)

(1) ÷ (2) $\tan\theta = \frac{a}{g}$

$\theta = \tan^{-1}\left(\frac{a}{g}\right)$

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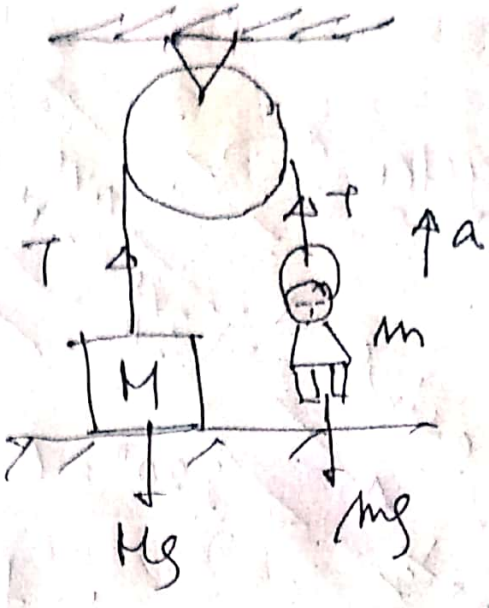
(8)

To lift M from the floor, cond^n is $T > Mg$ — (1)



Now for man,

$$T - mg = ma$$

or $T = m(g+a)$ — (2)

from (1)

$$\Rightarrow m(g+a) > Mg$$

$$\boxed{a > \left(\frac{M}{m} - 1\right) g}$$

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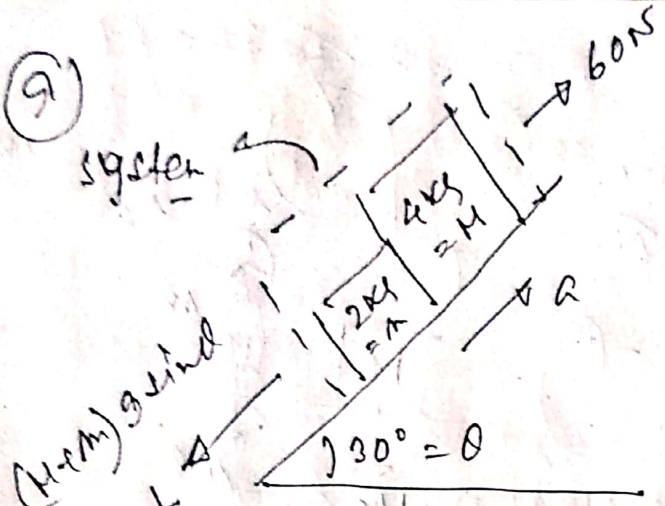
(9)



Let common accl of system (2kg+4kg) be 'a'.

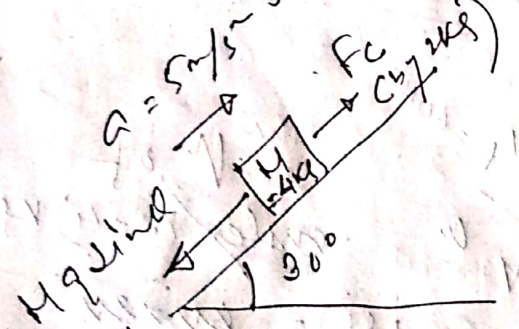
∴ for 2+4 = 6kg, by 2nd law of Newton $F_{net} = (M+m)a$

$60-30 = (4+2)a$

$a = \frac{30}{6} = 5 m/s^2$

Let contact force betn then be 'F_c'

Consider f.b.d of 4kg = M



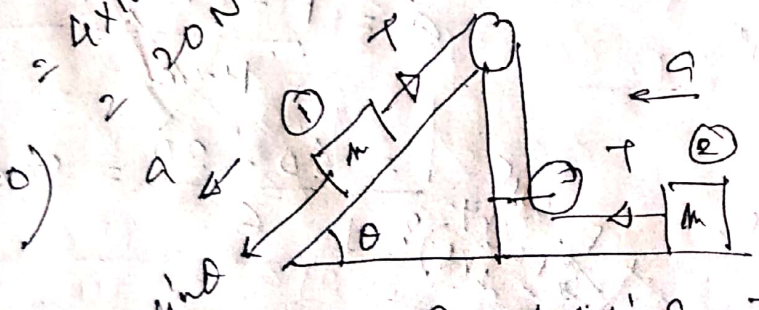
By 2nd law,

$F_c - 20 = Ma$

$F_c - 20 = 4 \times 5$

$\boxed{F_c = 40N}$

10)



for block ① $mg\sin\theta - T = ma$ — ①

& for block ② $T = ma$ — ②

① + ② $\cancel{m}g\sin\theta = 2\cancel{m}a \Rightarrow \boxed{a = \frac{g}{2}\sin\theta}$

∴ for ② $T = ma \Rightarrow \boxed{T = \frac{mg}{2}\sin\theta}$

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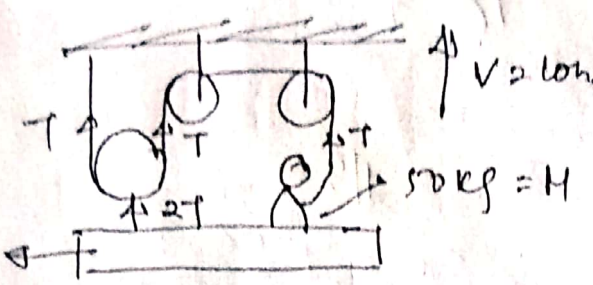
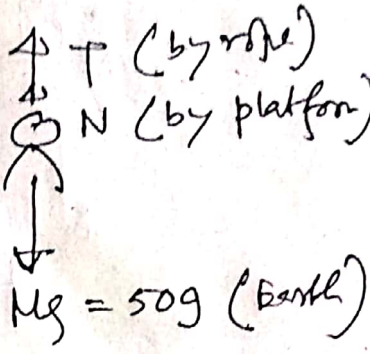
(12)



$\uparrow v = \text{const} \therefore a = 0.$

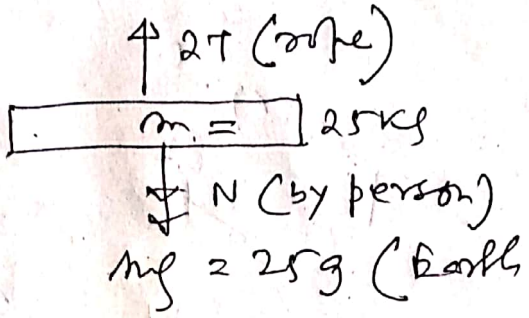
f.b.d of <u>person</u>



$$T + N = 50g \quad -(1)$$

$$N = 50g - T$$

f.b.d of <u>platform</u>



$$2T = N + 25g \quad -(2)$$

put N from (1) in (2)

$$2T = 50g - T + 25g$$

$$3T = 75g \Rightarrow T = 25g = 25 \times 10$$

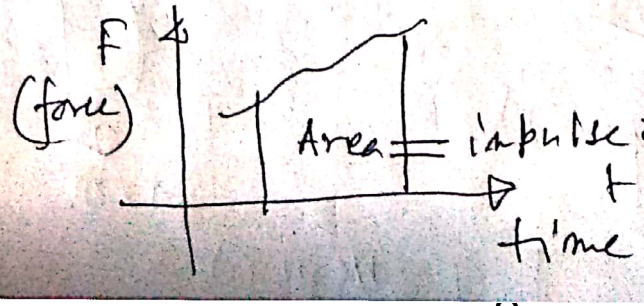
$$= \underline{\underline{250N}}$$

(14) <u>Very</u> <u>important</u> concept for JEE.

we have read, Impulse of a force = change in linear momentum. $(p_f - p_i)$

Now Impulse of a force F is

$$I = \int F\,dt = \text{area under } F\text{-}t \text{ graph.}$$



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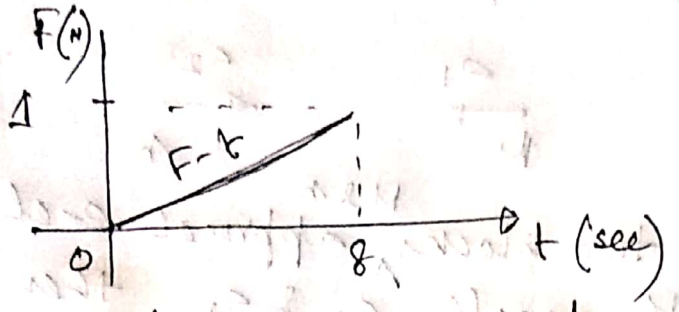
NLM JEE

Prob 14



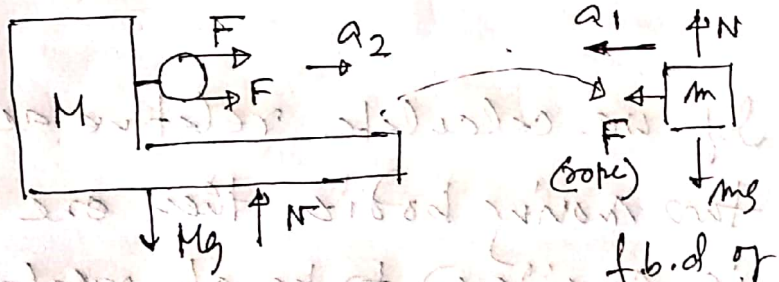
change in linear momentum betn $t=0$ to $t = 8$ sec

= impulse of Force (IN)

$= \int F\,dt$ = area under F-t graph

$= \frac{1}{2} \times 8 \times 1 = 4$ N-S

15



f.b.d of M

f.b.d of m

for M $\quad 2F = Ma_2$

$\therefore$ accl^n of M $\left| a_2 = \frac{2F}{M} \right|$

for m $\quad F = ma_1$

accl^n of m $\left| a_1 = \frac{F}{m} \right|$

Relative accl^n betn m & M, ie accl^n of m w.r.t M (considered to be rest)

$a = a_1 + a_2 = \frac{2F}{M} + \frac{F}{m} = F\left(\frac{2}{M} + \frac{1}{m}\right)$

Let 'm' hits the pulley in t sec.

$\therefore \; l = \frac{1}{2} a t^2 \Rightarrow t = \sqrt{\frac{2l}{a}} = \sqrt{\frac{2l}{F\left(\frac{2}{M}+\frac{1}{m}\right)}}$

$\sim t = \sqrt{\frac{2lMm}{(2m+M)F}}$.

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Concept required :

(1)

If two blocks M & m approach each other with accel a_1 & a_2, then accel of 1 block w.r.t other is $a = a_1 + a_2$

(2)

Let $(a_1 > a_2)$ If they move in same dir^n, their relative accel i.e accel of 1 block w.r.t other is $a = a_1 - a_2$

If we calculate relative accel betn. two moving bodies then one body is considered to be at rest & accel of other body is found relative to 1st body.

Then two body problem is reduced to a single body problem and hence it's easier to deal with motion of a single body.

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